Number Systems

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Common Number Systems

System	Base	Symbols	Used by humans ?	Used in computers?
Decimal	10	0, 1, … 9	Yes	No
Binary	2	0, 1	No	Yes
Octal	8	0, 1, … 7	No	No
Hexa-decimal	16	0, 1, … 9, A, B, … F	No	No

Quantities/Counting (2 of 3)

Decimal	Binary	Octal	Hexa-decimal
0	0	0	0
1	1	1	1
2	10	2	2
3	11	3	3
4	100	4	4
5	101	5	5
6	110	6	6
7	111	7	7

Quantities/Counting (2 of 3)

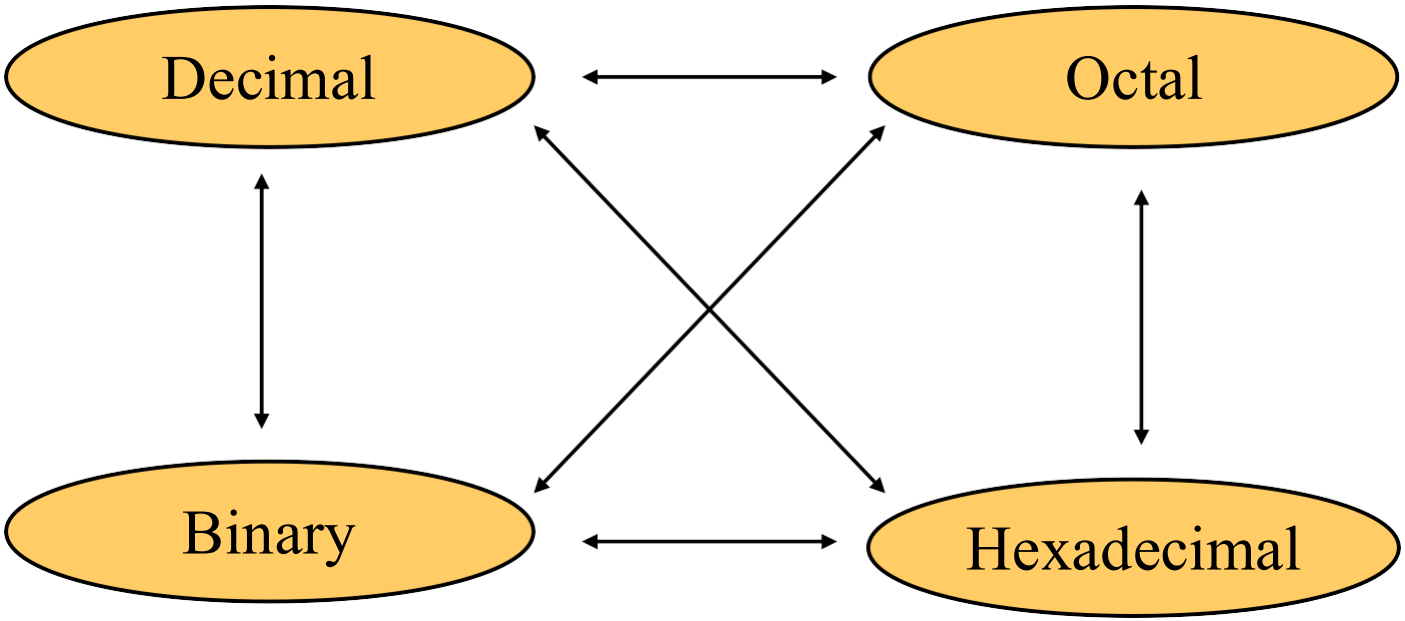
Decimal	Binary	Octal	Hexa-decimal
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Quantities/Counting (3 of 3)

Decimal	Binary	Octal	Hexa-decimal	
16	10000	20	10	
17	10001	21	11	
18	10010	22	12	
19	10011	23	13	
20	10100	24	14	
21	10101	25	15	
22	10110	26	16	
23	10111	27	17	Etc.

Conversion Among Bases

- The possibilities:



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Quick Example

$$25_{10} = 11001_2 = 31_8 = 19_{16}$$

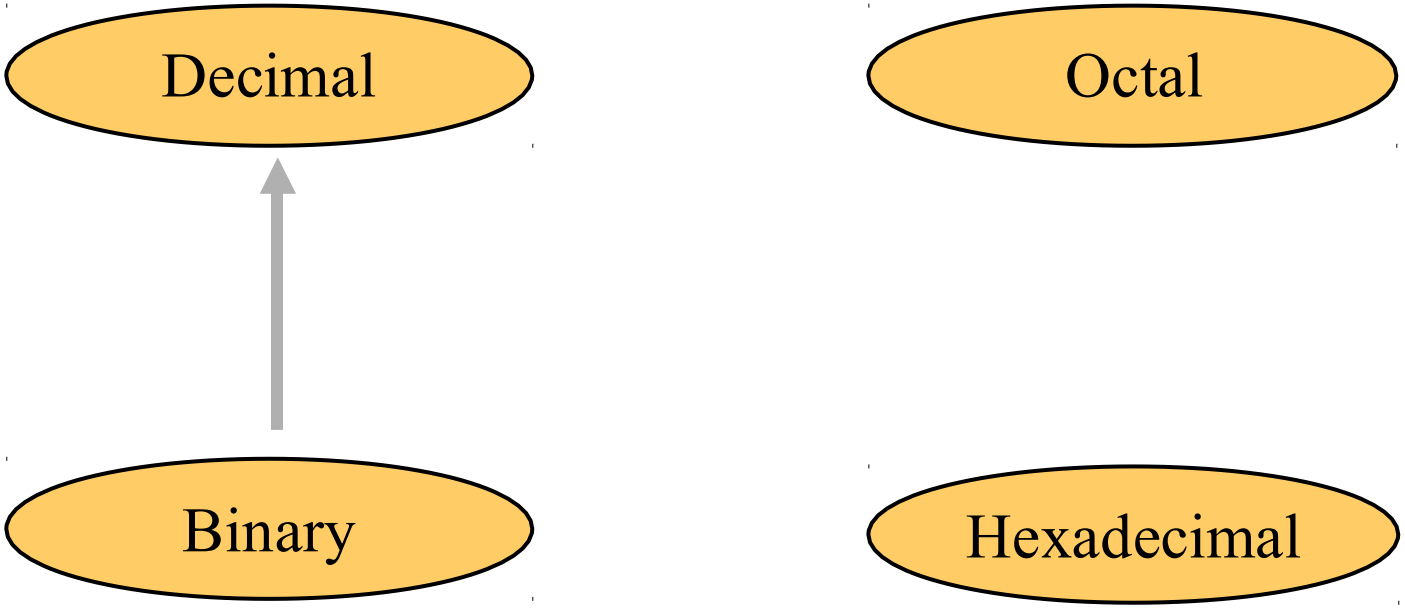
Base

Weight

$$125_{10} \Rightarrow \quad \begin{array}{lcr} 5 \times 10^0 & = & 5 \\ 2 \times 10^1 & = & 20 \\ 1 \times 10^2 & = & \underline{100} \\ & & 125 \end{array}$$

Base

Binary to Decimal



Binary to Decimal

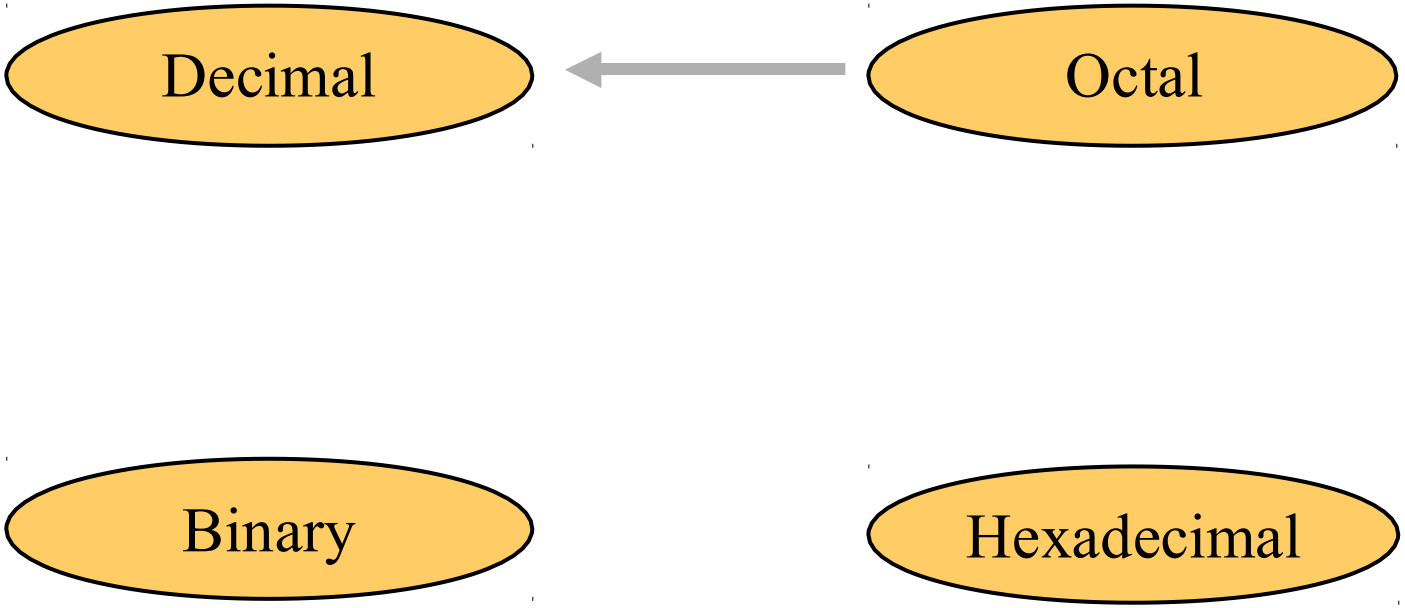
- Technique
  - Multiply each bit by 2^n, where n is the "weight" of the bit
  - The weight is the position of the bit, starting from 0 on the right
  - Add the results

Example

101011_2 =>

$$
\begin{array}{lcl}
1 \times 2^0 & = & 1 \\
1 \times 2^1 & = & 2 \\
0 \times 2^2 & = & 0 \\
1 \times 2^3 & = & 8 \\
0 \times 2^4 & = & 0 \\
1 \times 2^5 & = & \underline{32} \\
 & & 43_{10}
\end{array}
$$

Octal to Decimal



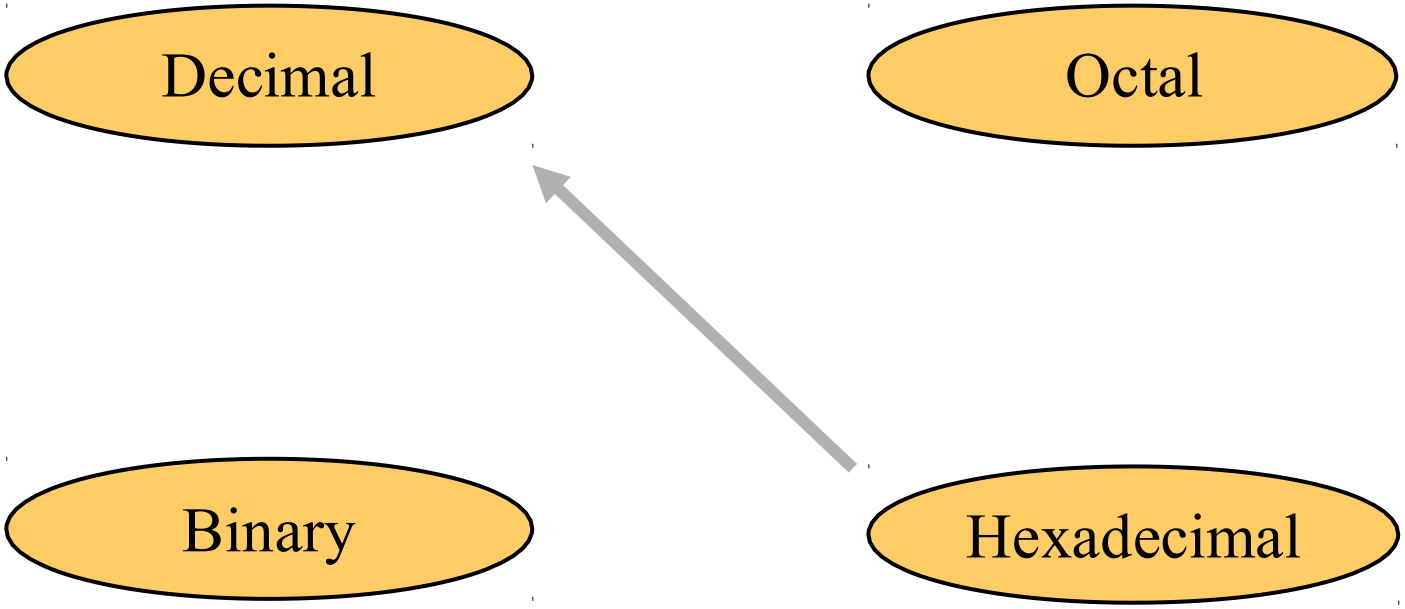
Octal to Decimal

- Technique
  - Multiply each bit by 8^n, where n is the "weight" of the bit
  - The weight is the position of the bit, starting from 0 on the right
  - Add the results

Example

$$
\begin{array}{lrcl}
724_8 \Rightarrow & 4 \times 8^0 & = & 4 \\
 & 2 \times 8^1 & = & 16 \\
 & 7 \times 8^2 & = & \underline{448} \\
 & & & 468_{10}
\end{array}
$$

Hexadecimal to Decimal



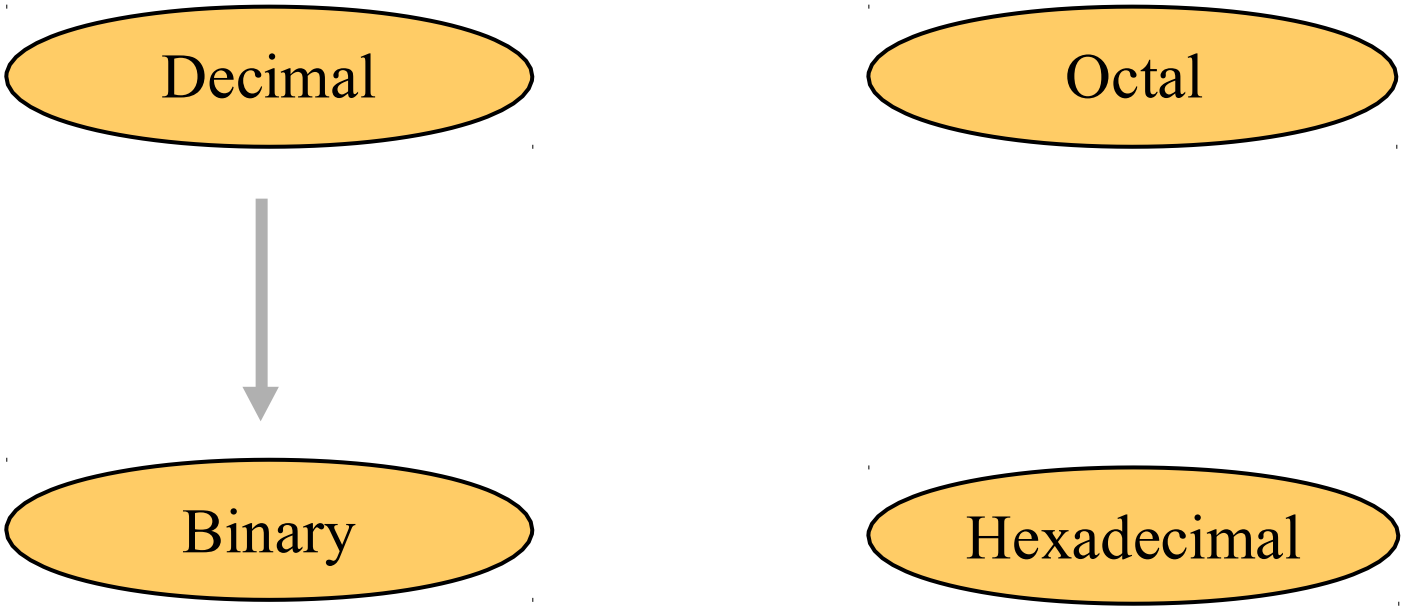
Hexadecimal to Decimal

- Technique
  - Multiply each bit by 16^n, where n is the "weight" of the bit
  - The weight is the position of the bit, starting from 0 on the right
  - Add the results

Example

$$
\begin{array}{lrcrcrcr}
ABC_{16} \Rightarrow & C & \times & 16^0 & = & 12 & \times & 1 & = & 12 \\
 & B & \times & 16^1 & = & 11 & \times & 16 & = & 176 \\
 & A & \times & 16^2 & = & 10 & \times & 256 & = & \underline{2560} \\
 & & & & & & & & & 2748_{10}
\end{array}
$$

Decimal to Binary



Decimal to Binary

- Technique
  - Divide by two, keep track of the remainder
  - First remainder is bit 0 (LSB, least-significant bit)
  - Second remainder is bit 1
  - Etc.

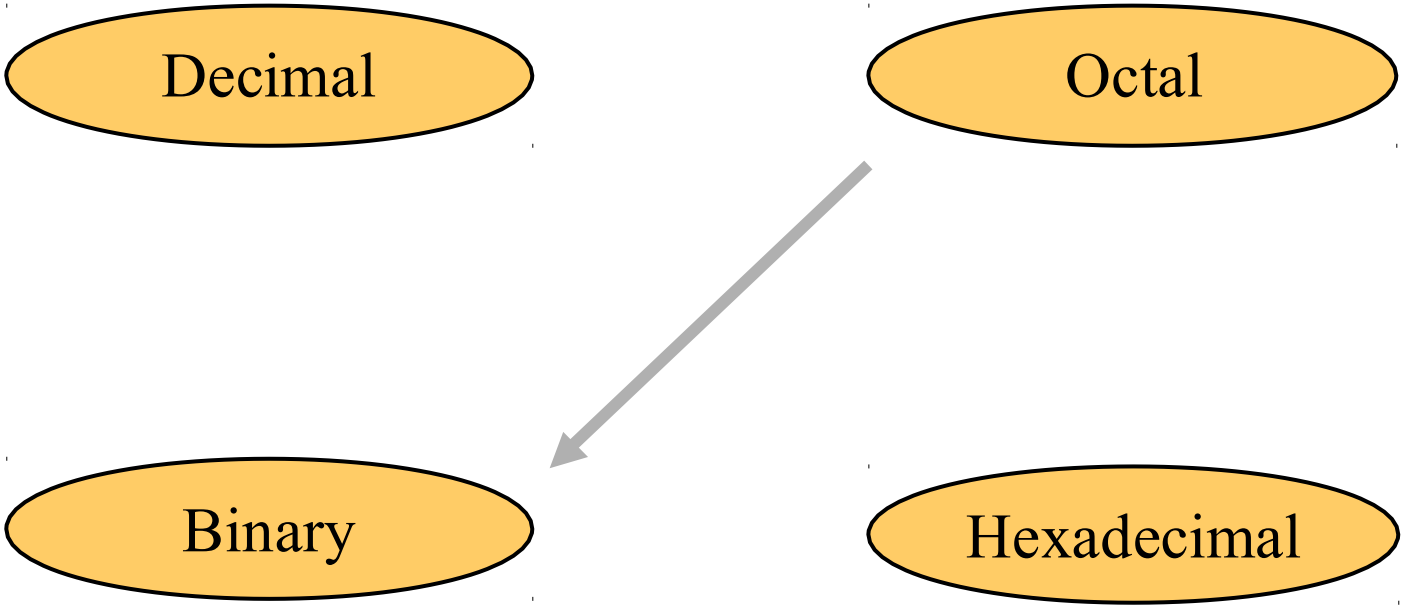
Example

$125_{10} = ?_2$

2	125	
2	62	1
2	31	0
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

$125_{10} = 1111101_2$

Octal to Binary



Octal to Binary

- Technique
  - Convert each octal digit to a 3-bit equivalent binary representation

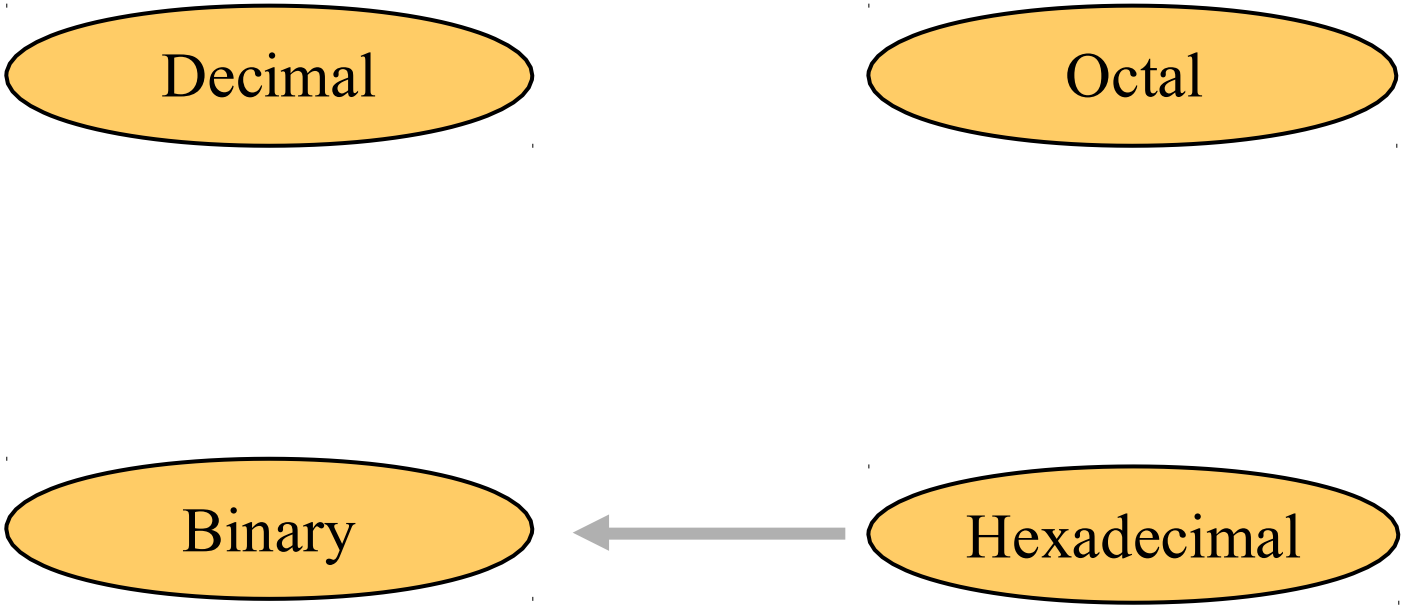
Example

$705_8 = ?_2$

7	0	5
↓	↓	↓
111	000	101

$705_8 = 111000101_2$

Hexadecimal to Binary



Hexadecimal to Binary

- Technique
  - Convert each hexadecimal digit to a 4-bit equivalent binary representation

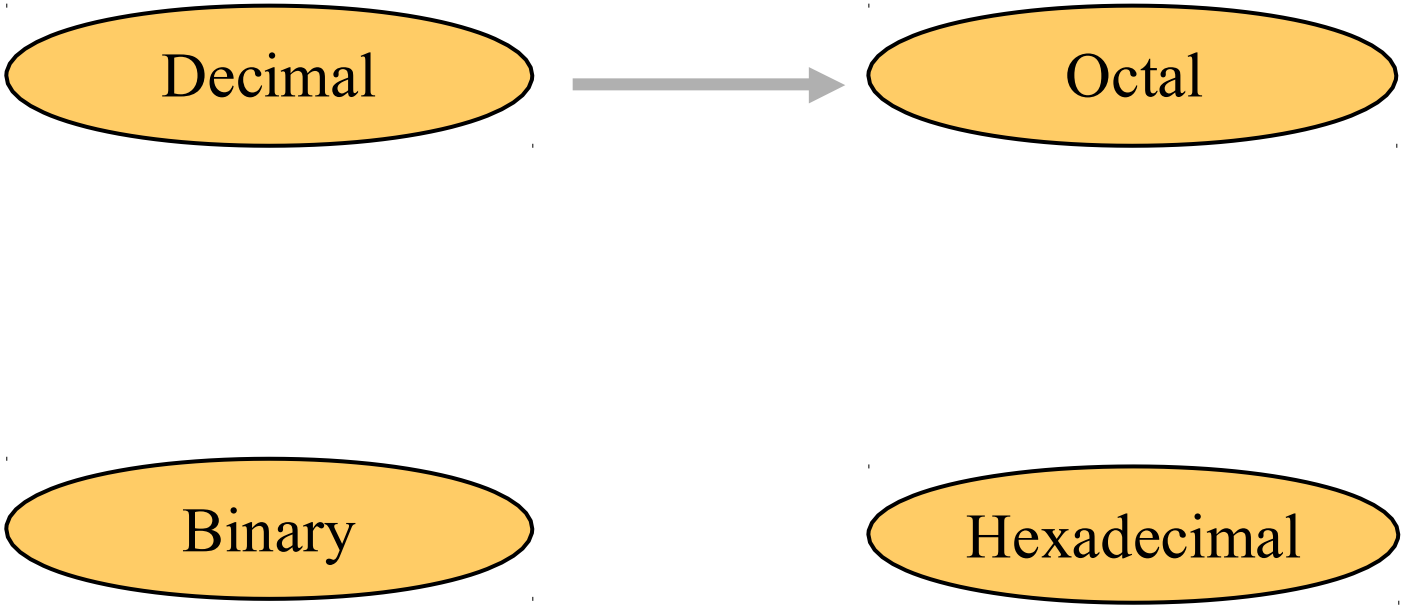
Example

$10AF_{16} = ?_2$

1	0	A	F
↓	↓	↓	↓
0001	0000	1010	1111

$10AF_{16} = 0001000010101111_2$

Decimal to Octal



Decimal to Octal

- Technique
  - Divide by 8
  - Keep track of the remainder

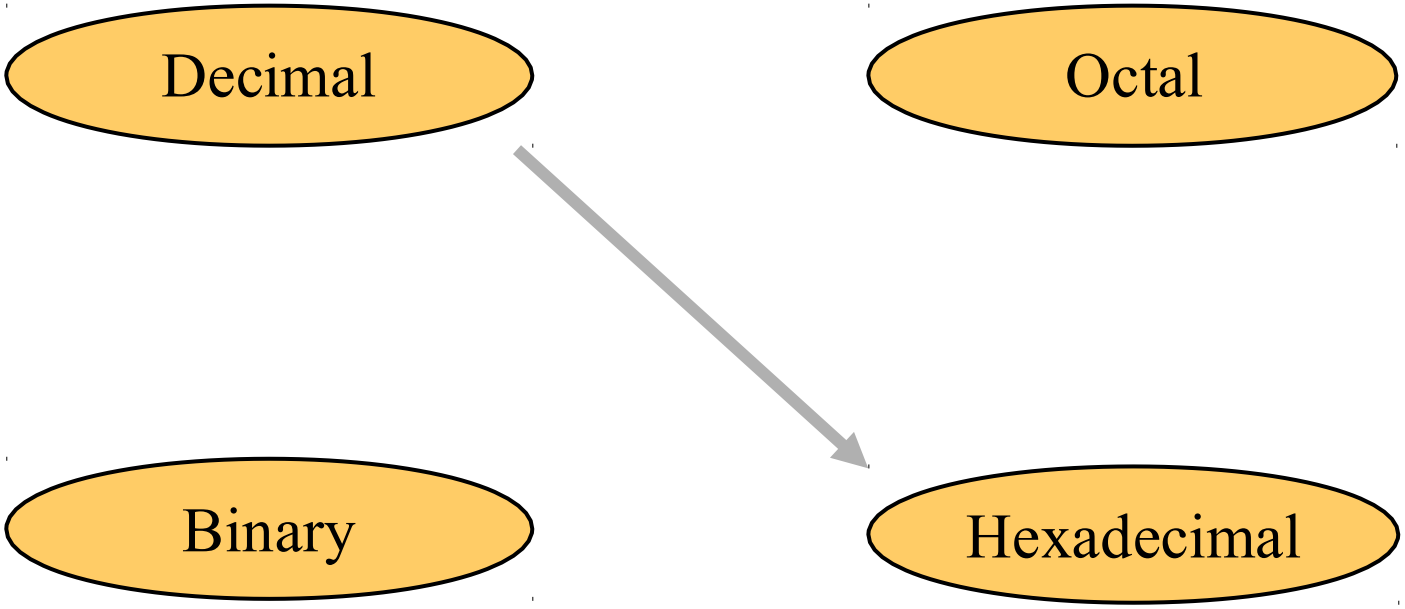
Example

$1234_{10} = ?_8$

```
8   1234
8    154    2
8     19    2
8      2    3
        0   2
```

$1234_{10} = 2322_8$

Decimal to Hexadecimal



Decimal to Hexadecimal

- Technique
  - Divide by 16
  - Keep track of the remainder

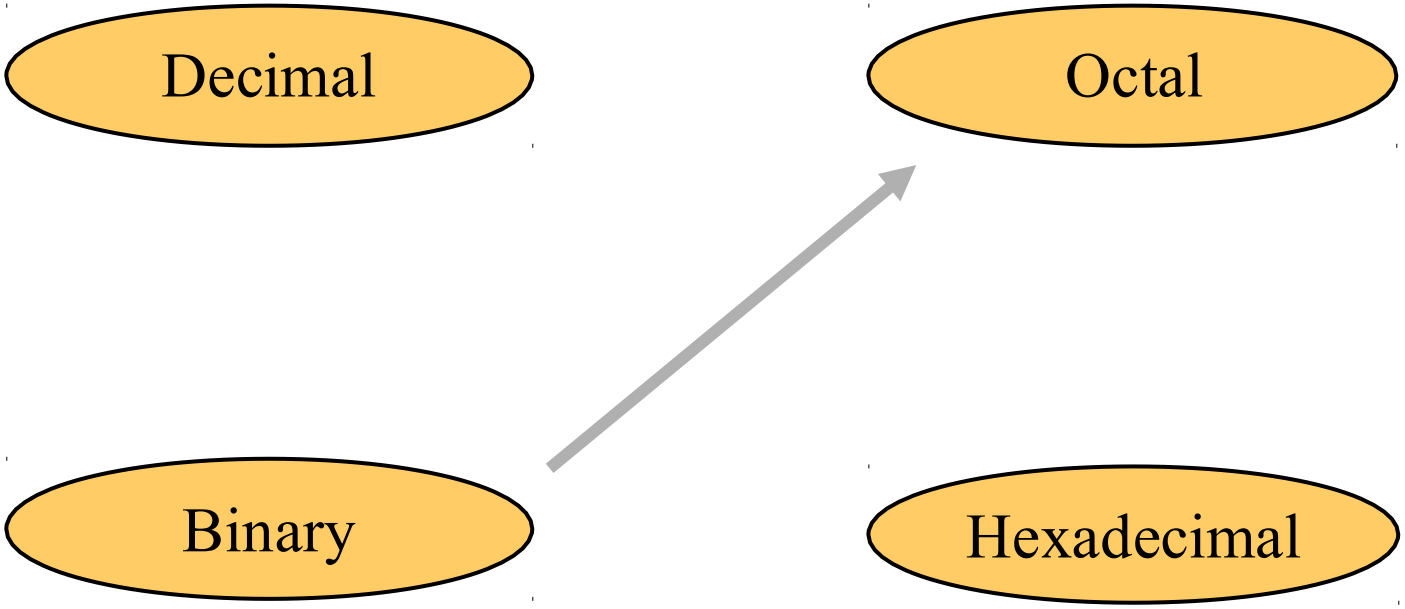
Example

$1234_{10} = ?_{16}$

16	1234	
16	77	2
16	4	13 = D
	0	4

$1234_{10} = 4D2_{16}$

Binary to Octal



Binary to Octal

- Technique
  - Group bits in threes, starting on right
  - Convert to octal digits

Example

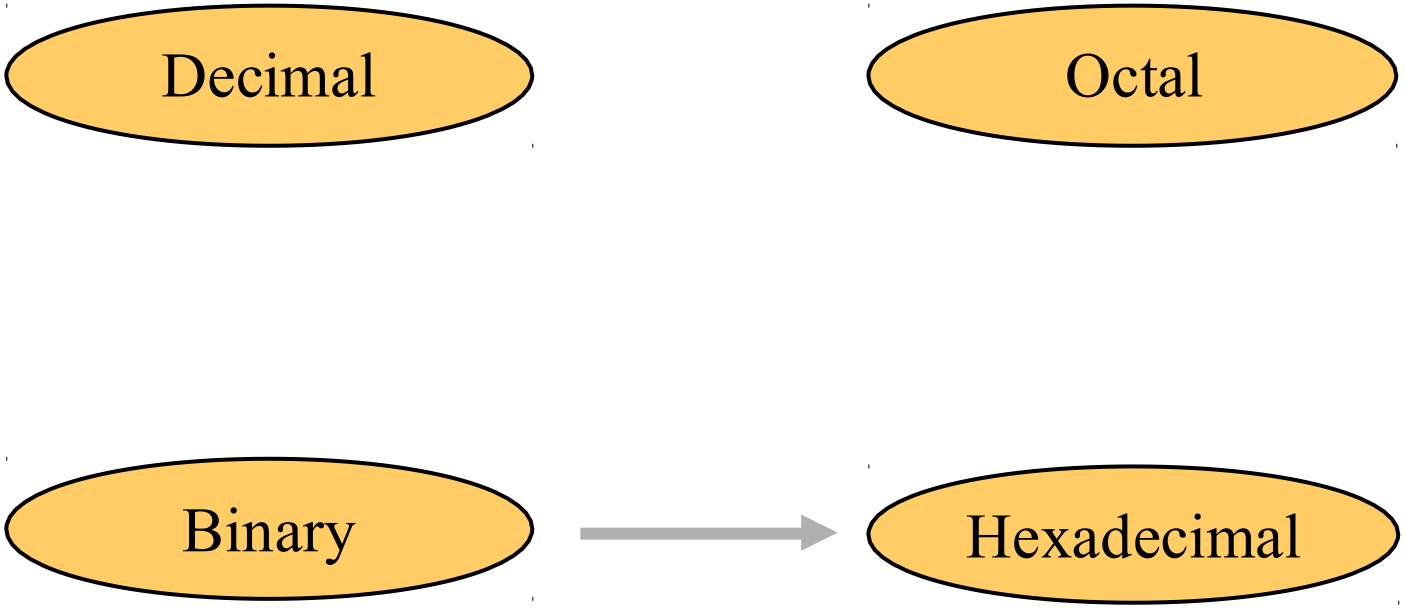
$1011010111_2 = ?_8$

1 011 010 111

1 3 2 7

$1011010111_2 = 1327_8$

Binary to Hexadecimal



Binary to Hexadecimal

- Technique
  - Group bits in fours, starting on right
  - Convert to hexadecimal digits

Example

$1010111011_2 = ?_{16}$

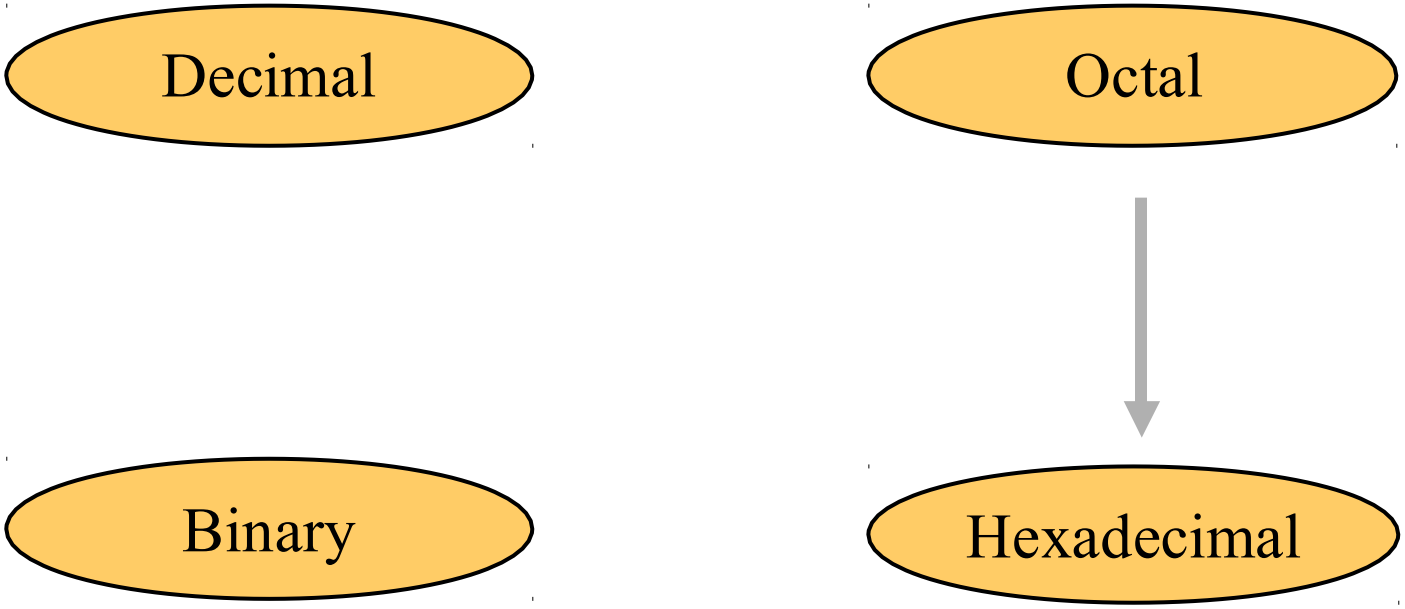
10 1011 1011

↓ ↓ ↓

2 B B

$1010111011_2 = 2BB_{16}$

Octal to Hexadecimal



Octal to Hexadecimal

- Technique
  - Use binary as an intermediary

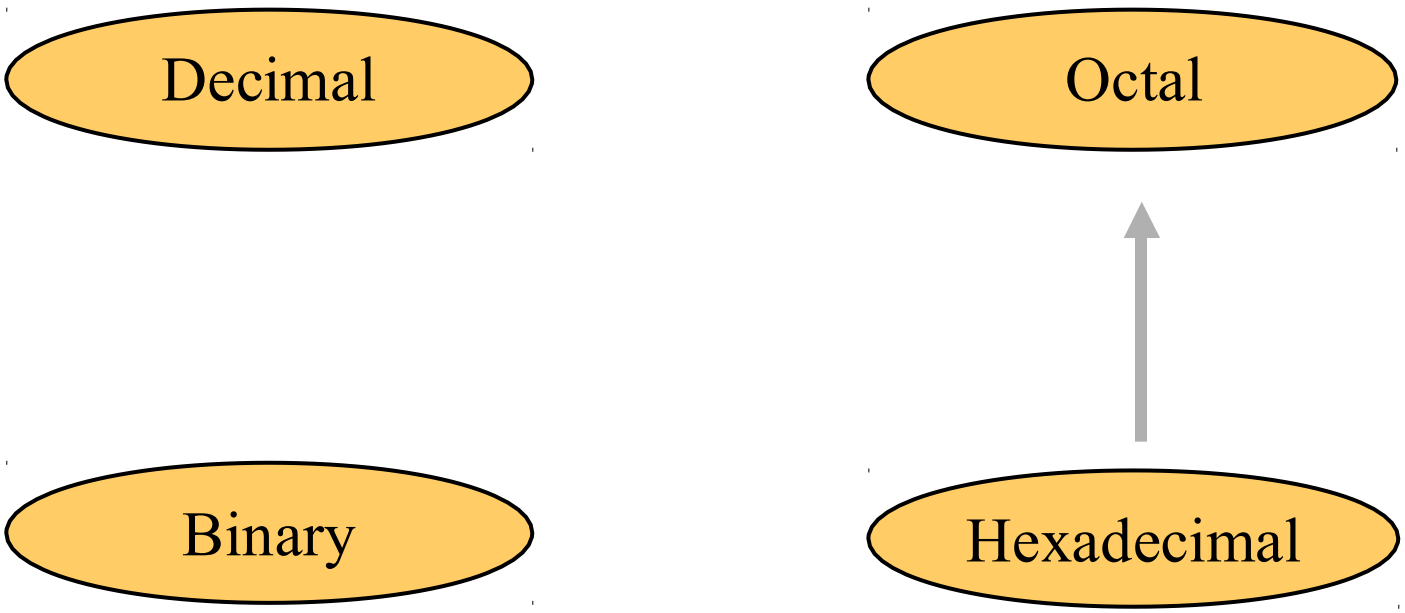
Example

$1076_8 = ?_{16}$

1	0	7	6
↓	↓	↓	↓
001	000	111	110
2	3		E

$1076_8 = 23E_{16}$

Hexadecimal to Octal



Hexadecimal to Octal

- Technique
  - Use binary as an intermediary

Example

$1F0C_{16} = ?_8$

1	F	0	C
0001	1111	0000	1100
1	7	4	1	4

$1F0C_{16} = 17414_8$

Exercise – Convert ...

Decimal	Binary	Octal	Hexa-decimal
33			
	1110101		
		703	
			1AF

Don't use a calculator!

Skip answer

Answer

Exercise – Convert …

Decimal	Binary	Octal	Hexa-decimal
	100001	41	21
117		165	75
451	111000011		1C3
431	110101111	657	1AF

Common Powers (1 of 2)

- Base 10

Power	Preface	Symbol	Value
10^{-12}	pico	p	.000000000001
10^{-9}	nano	n	.000000001
10^{-6}	micro	μ	.000001
10^{-3}	milli	m	.001
10^{3}	kilo	k	1000
10^{6}	mega	M	1000000
10^{9}	giga	G	1000000000
10^{12}	tera	T	1000000000000

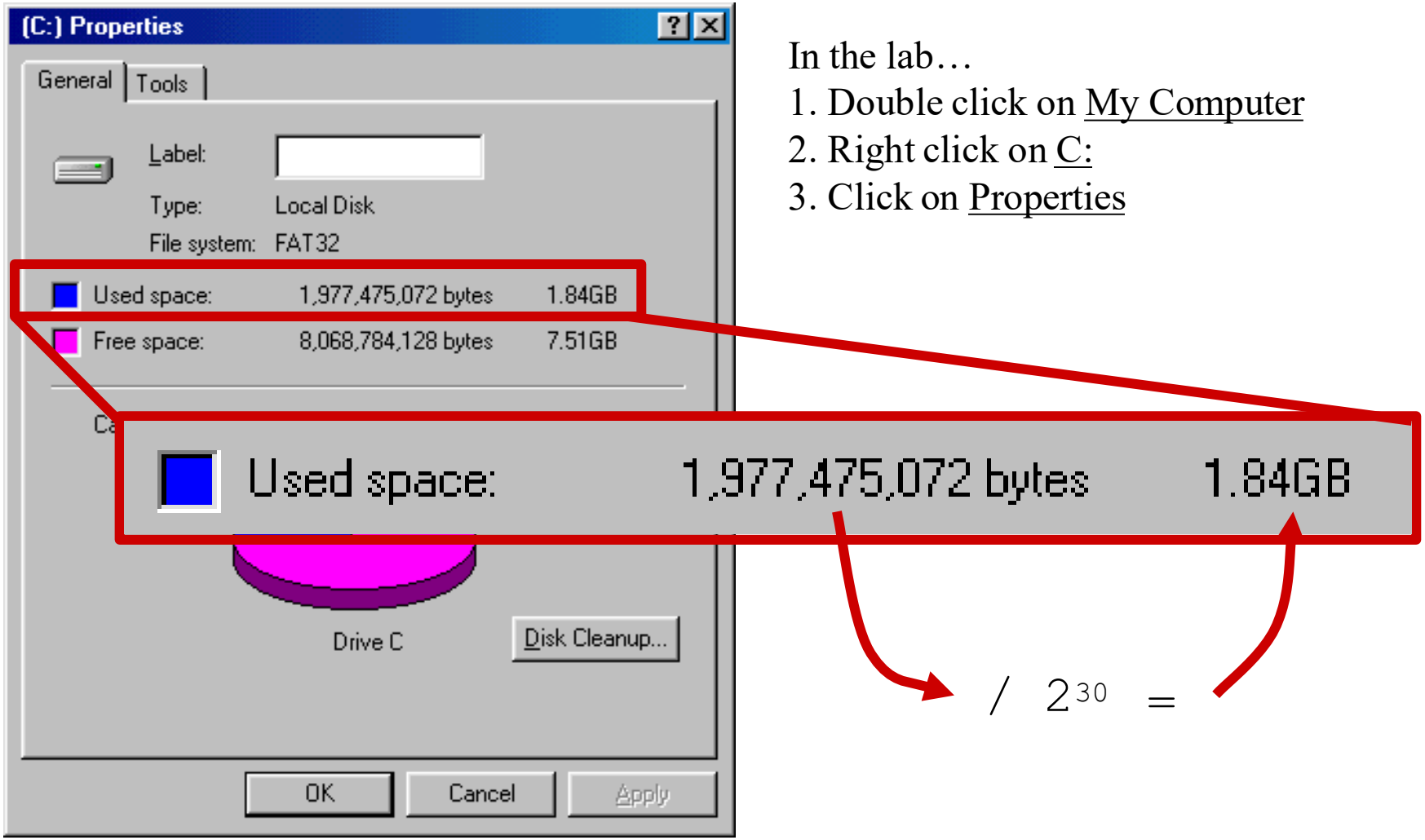
Common Powers (2 of 2)

- Base 2

Power	Preface	Symbol	Value
2^{10}	kilo	k	1024
2^{20}	mega	M	1048576
2^{30}	Giga	G	1073741824

- What is the value of “k”, “M”, and “G”?
- In computing, particularly w.r.t. <u>memory</u>, the base-2 interpretation generally applies

Example



In the lab…

1. Double click on My Computer
2. Right click on C:
3. Click on Properties

Thank you